

MyDatabaseTheory

Database Theory – Practical Task

1. Explain the difference between data and information.

Data is raw, unprocessed facts and figures. On their own, they may not tell us very much. Information is data that has been organised and given some context, which makes it useful and easier to understand. For example, “21” is just data, but if we say “Andy is 21 years old”, it becomes information because we now know what the number means.

2. What is metadata?

Metadata is information that describes other data. It helps us understand what the data represents and how it should be used. Examples of metadata include labels, units of measurement, and the period of time that the data covers.

3. What is a DBMS, and what are its advantages?

A Database Management System (DBMS) is software that is used to manage a database. It controls how users and applications access and work with the data. A DBMS makes it easier to access data while also helping to keep it secure. It can also improve the way data is organised and integrated, make it easier to update or retrieve information, and help users get better information for making decisions.

4. Compare and contrast operational databases with analytical databases and provide an example of each.

Operational databases are used for the everyday activities of an organisation. They contain current, detailed information and are designed to handle frequent updates and data retrieval. An example would be a database used to process customer orders or manage stock.

Analytical databases are mainly used to analyse data and create reports. They normally contain historical and aggregated information and are designed to handle more complex queries. A data warehouse is an example of an analytical database.

5. Explain the types of data and use cases where NoSQL databases are most effective.

There are three main types of data where NoSQL databases are most effective. Structured data is organised and neat, presented in rows and columns, such as in an Excel sheet. Semi-structured data has some organisation but is more flexible; JSON and XML are examples. Unstructured data does not have a set format and can include things such as documents, images and videos.

NoSQL databases are useful when an application needs to work with large amounts of unstructured or semi-structured data. They are also useful when a flexible data structure and scalability are important.

6. Which DBMS does not require server configuration, and what are the advantages of using it?

The DBMS that does not require server configuration is SQLite. It is lightweight and portable. The whole database is just one file. You can easily copy, move, or back it up. It's also very small, so it works well on mobile phones and embedded devices.

7. Explain ACID properties in the context of a DBMS.

ACID stands for Atomicity, Consistency, Isolation, and Durability. Atomicity means that a transaction is treated as a whole. If any part of the transaction fails, the database rolls back all changes made during that transaction. Consistency means that the database remains correct after a transaction. Isolation means that transactions happening at the same time should not interfere with one another. Durability means that once a change has been saved, it remains saved and can be recovered if the system crashes.